

**5** Obesity increases the risk of cardiovascular disease (CVD).

One way to reduce obesity is to lose weight by changing eating habits.

(a) Name **two** obesity indicators used by scientists.

(1)

1 .....

2 .....

(b) Glucomannan is a dietary supplement claimed to aid weight loss.

Glucomannan is a branched polysaccharide similar in structure to amylopectin.

(i) Which glycosidic bonds are responsible for the branching in glucomannan?

(1)

☐ **A** 1-4 only

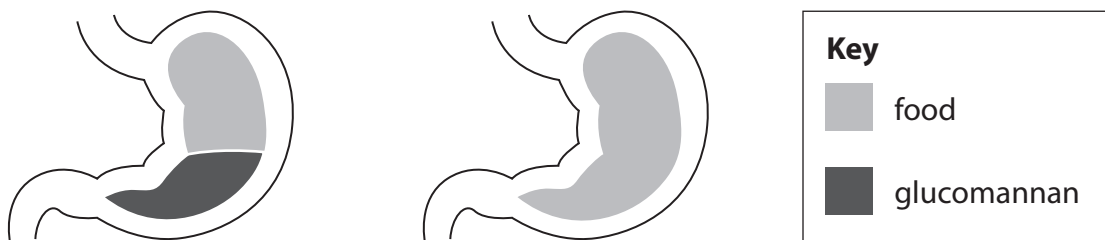
☐ **B** 1-6 only

☐ **C** both 1-4 and 1-6

☐ **D** neither 1-4 nor 1-6

(ii) In the presence of water, glucomannan swells to form a semi-solid gel.

The diagram shows a stomach with glucomannan present and a stomach without glucomannan.



Suggest how glucomannan aids weight loss.

(1)

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(iii) If glucomannan could be digested it would cause a gain in weight.

Explain why glucomannan could cause a gain in weight.

(2)

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- (c) A study compared weight loss in two groups of women.

One group of women had a low-fat diet and the other group of women had a very low-carbohydrate diet.

- (i) The table shows the results of this study.

| Number of weeks<br>on the diet | Mean body mass of the group of women / kg |                                            |
|--------------------------------|-------------------------------------------|--------------------------------------------|
|                                | Group on the<br>low-fat diet              | Group on the very<br>low-carbohydrate diet |
| 0                              | 92.5                                      | 91.1                                       |
| 2                              | 90.6                                      | 88.3                                       |
| 4                              | 89.9                                      | 87.5                                       |
| 6                              | 89.0                                      | 85.6                                       |
| 8                              | 88.8                                      | 84.5                                       |
| 10                             | 88.3                                      | 84.1                                       |
| 12                             | 88.2                                      | 83.0                                       |

Many studies claim that a low-carbohydrate diet can result in two to three times as much weight loss as a low-fat diet.

Determine the extent to which this study supports this claim.

(3)

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(ii) Very low-carbohydrate diets may increase cardiovascular risk factors.

Give **two** factors, other than weight loss, that should have been monitored in this study.

(2)

1 .....

2 .....

(Total for Question 5 = 10 marks)

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